Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	1260	**2**	**M1** for $15 \times 54 + 25 \times 18$
1(b)	38 800	**2**	**M1** for $37054 \div \left(1 - \frac{4.5}{100}\right)$ oe
1(c)(i)	15 : 12 : 28	**2**	**M1** for correct attempt to find a common multiple for the women oe
1(c)(ii)	216	**3**	**M2** for $224 \div$ *their* $28 \times$ *their* $(15 + 12)$ or **M1** for $224 \div$ *their* 28
1(d)	55.25	**2**	**M1** for 8 + 0.5 or 6 + 0.5 seen
1(e)	156 or 156.3…	**2**	**M1** for $\left(1 + \frac{1.5}{100}\right)^{30}$
2(a)(i)	triangle with vertices at (–2, –1) (–8, –1) (–2, –5)	**2**	**B1** for correct reflection in $y = x$
2(a)(ii)	triangle with vertices at (–1, –1) (–1, –7) (3, –7)	**2**	**B1** for translation by $\begin{pmatrix} k \\ -9 \end{pmatrix}$ or $\begin{pmatrix} -2 \\ k \end{pmatrix}$
2(b)(i)	Enlargement [centre] (–7, 8) [sf] ½	**3**	**B1** for each
2(b)(ii)	Rotation [centre] (0, 0) 90° clockwise oe	**3**	**B1** for each
3(a)	correct diagram	**4**	**B1** for median line correctly drawn at 148 **B1** for 105 soi **B1** for whisker at 159 soi
3(b)	6.48	**3**	**M1** for $(5 \times 8) + (6 \times 2) + (12 \times 7) + \ldots$ **M1dep** for *their* $\sum fx \div$ *their* $(8 + 2 + 12 + 2 + 0 + 1)$
4(a)	$m \geq 3.4$ oe final answer	**2**	**M1** for $12 + 5 \leq 8m - 3m$ or better or $3m - 8m \leq -5 - 12$ or better

Question	Answer	Marks	Partial Marks
4(b)	$x = -0.75$ oe	3	**M1** for $15(2x+5)=14(3-x)$ **B1** for $30x+75=42-14x$ or better
4(c)	$3x^2-16x-35[=0]$ or $3y^2-8y-51[=0]$	M3	**M1** for $x^2+2(4-x)^2=67$ or $(4-y)^2+2y^2=67$ seen **B1** for $16-8x+x^2$ or $16-8y+y^2$
	$(3x+5)(x-7)$ [= 0] or $(3y-17)(y+3)$[= 0]	M1	or for correct factors for *their* equation or for correct use of quadratic formula or completing the square for *their* equation
	$x = 7,\ y = -3$ $x=-\frac{5}{3},\ y=5\frac{2}{3}$	B2	**B1** for $x = 7,\ x=-\frac{5}{3}$ or for $y = -3,\ y=5\frac{2}{3}$ or for a correct pair of x and y values
5(a)	$(4x-5)(x+3)+(x+1)(x-3)=342$ or $2x(4x-5)-(3x-6)(x-3)=342$	M2	**M1** for $(4x-5)(x+3)$ or $(x+1)(x-3)$ or for $2x(4x-5)$ or $(3x-6)(x-3)$
	$4x^2+12x-5x-15$ oe and $x^2+x-3x-3$ oe seen OR $8x^2-10x$ and $3x^2-15x+18$ seen	M2	**M1** for each
	$5x^2+5x-18=342$ leading to $x^2+x-72=0$	A1	no errors or omission
5(b)	$(x+9)(x-8)$	M2	**B1** for $(x+a)(x+b)$ where $ab = -72$ or $a + b = 1$ and a, b are integers
	8, –9	B1	
5(c)	86	2	**FT** for $12\times$*their* $x-10$ (x positive) **B1** for any one of 27, 11, 16 seen or for $2x+2x+4x-5+4x-5$ oe or better soi
5(d)	22.2 or 22.16 to 22.17	2	**M1** for tan = $\frac{11}{27}$ or $\frac{\textit{their } x+3}{4\times\textit{their } x-5}$
6(a)(i)	29.5 or 29.50…	4	**M2** for $\frac{11^2+5.3^2-6.9^2}{2\times11\times5.3}$ or **M1** for $6.9^2 = 11^2+5.3^2-2\times11\times5.3\cos x$ **A1** for 0.87[0…] oe